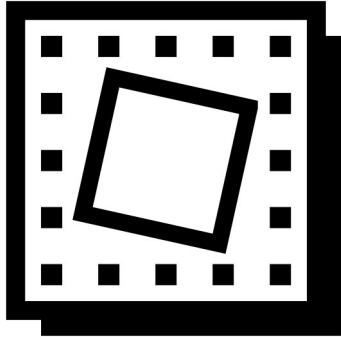




PAPLITZ

OPEN SOURCE • SPATIAL RESEARCH

Digital Pedagogy and Real-Time Analytical Feedback for Technical Sketching and 3D Perspective Training in Industrial Design

AUTHORSHIP:

Lázaro Guerrero Losada

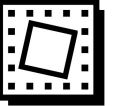
B.Sc. in Industrial Design Engineering and Product Development (UMA)

M.Sc. Student in Industrial Engineering (UMA)

PROJECT INFO:

Project Web: <https://paplitz.vercel.app/>

Email: lazaroguerreroslada@uma.es



01

Context, Motivation & Solution

The sketching paradox in Industrial Design: high demand without explicit methodology.

03

Project Objectives

Democratizing learning through open-source software and deliberate practice drills.

02

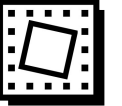
Pedagogical Influences & Digital Transition

From canonical literature (Koos Eissen, Alphonso Dunn) and Daromeon, to the iPad era.

04

Instructional Methodology

Progressive structuring of perspective concepts and spatial drawing skills.



ACADEMIC PROBLEM STATEMENT

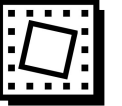
The Myth of "Spontaneous Sketching"

- **Lack of explicit methodology:** In industrial design degree programs, freehand sketching is rarely taught systematically step by step, despite being constantly required in design studio courses.
- **Lack of objective metrics:** How does a student know if their vanishing convergence is accurate? On physical paper, feedback is virtually nonexistent without one-on-one expert tutoring.
- **Self-taught isolation:** Students are left to practice on their own. Without immediate feedback, incorrect perspective habits become unconsciously ingrained.
- **Frustration & dropout:** Steep learning curves cause many students to avoid freehand sketching altogether, relying prematurely on CAD modeling.

01. PAPTITZ

Background & Motivation: The Degree Paradox

The origin of a real pedagogical need experienced in Industrial Design classrooms

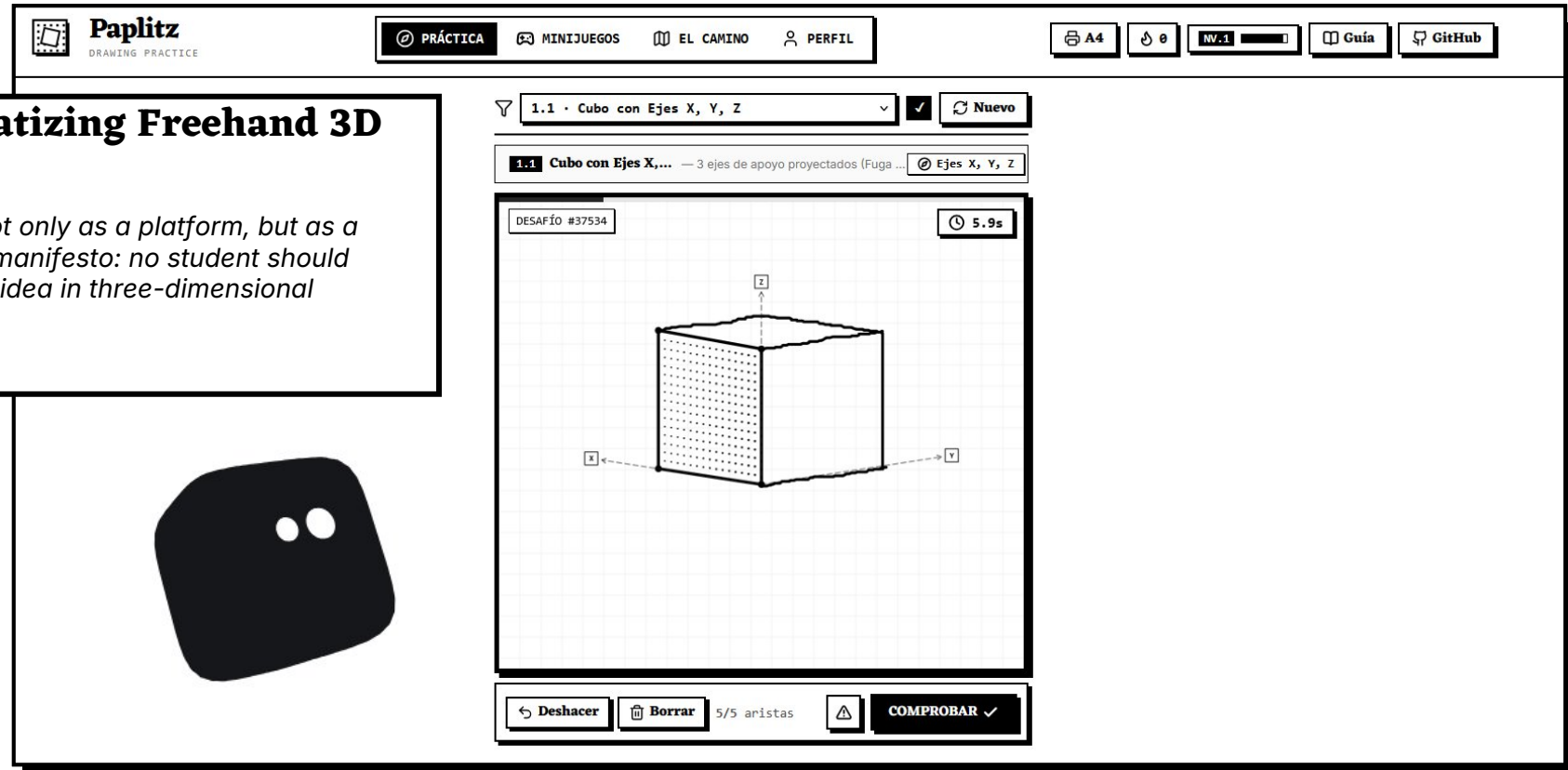


THE SOLUTION

Paplitiz: Democratizing Freehand 3D Sketching

Paplitiz was conceived not only as a platform, but as a technical and emotional manifesto: no student should feel unable to convey an idea in three-dimensional space.

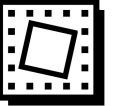
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01. PAPTITZ

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THE SOLUTION

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MISSION

To provide an open digital motor-skill trainer that breaks down conic perspective into measurable, gamified exercises.

VISION

To develop an open-source standard for design, art, and architecture schools, bringing together students and faculty.

VALUES

Universal access, learning gamification, complete privacy, and analytical rigor

02.

REFERENCES

Pedagogical Influences & The Transition to the Digital Canvas

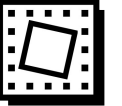
*From canonical literature to modern digital input
devices*



02. REFERENCES

Pedagogical Influences & The Transition to the Digital

Canvas *From traditional literature to modern digital input devices*



- **"Sketching: The Basics"**
(Koos Eissen & Roselien Steur)

Provides theoretical rigor: elliptical axis calculation, horizon lines, converging vanishing planes, and volumetric sectioning.

- **"Pen and Ink Drawing"**
(Alphonso Dunn)

Essential guide for stroke control, line weight hierarchy, hatching / cross-hatching, and monochrome ink lighting discipline.

- **Daromeon Interactive**

Pioneering inspiration for digital perspective micro-exercises in a web environment

ENVIRONMENTS

OVERCOMING THE "PAPER RESISTANCE"

**Paper as an
unscalable
medium**

**The rise of the
iPad and Active
Styluses**

**Instantaneo
us
evaluation**

03.

OBJECTIVES

Project Objectives & Paplitz Development

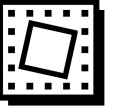
*Technical, pedagogical, and community impact
goals*



03. OBJECTIVES

Project Objectives & Paplitz Development

Technical, pedagogical, and community impact goals



GENERAL RESEARCH OBJECTIVE

To develop, validate, and implement an open-source interactive training ecosystem for accelerated spatial muscle memory and geometric self-correction in technical perspective sketching, accessible to any student without financial or hardware barriers.

SUB-OBJECTIVE 1: PLATFORM

Open-Source Web Platform Development

Build an accessible online platform for all users to practice and refine sketching technique.

SUB-OBJECTIVE 2: METHODOLOGY

Segmented Progressive Pedagogy

Deconstruct spatial complexity into thematic units and incremental scaffolded lessons.

SUB-OBJECTIVE 3: USER EXPERIENCE

Gamification & Feedback

Implement positive reinforcement mechanics that eliminate fear of failure and encourage deliberate daily practice.

SUB-OBJECTIVE 4: EDUCATIONAL ECOSYSTEM

Open Community (Students + Faculty)

Foster collaboration in design classrooms so educators and students can audit and contribute new exercises.

04.

METHODOLOGY

Instructional Methodology: The Learning Path

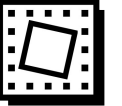
Curricular segmentation inspired by micro-learning and spaced reinforcement



04. METHODOLOGY

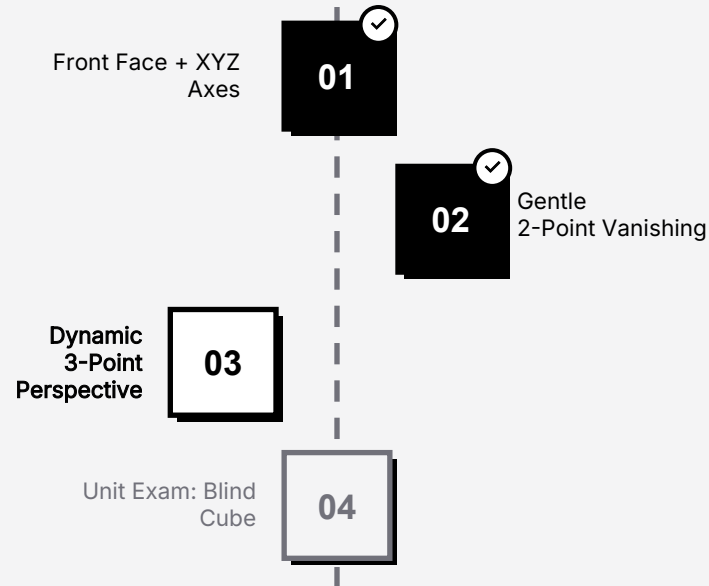
Instructional Methodology: The Learning Path ("The Path")

Curricular segmentation inspired by micro-learning and spaced reinforcement



WEB ZIGZAG LEARNING PATH

Unit 01: Foundations of Perspective



INSTRUCTIONAL DESIGN PRINCIPLES

Atomized Deconstruction of Spatial Drawing

1. Milestone Deconstruction

Beginners are not asked to draw a car or hair dryer; the challenge is isolated to the core primitive: the orthogonal cube.

2. Scaffolding

Assisted initial phase followed by progressive removal of guidelines until reaching the unassisted expert phase.

3. Reinforcement System

Every completed stroke receives instantaneous percentage scoring, clearly showing spatial progress.

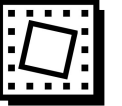
4. Placement & Level Exams

Once students achieve sufficient mastery, a placement challenge verifies competence to unlock advanced modules.

04. METHODOLOGY

The Interactive Loop: Challenge, Stroke & Evaluation

Real-time captures of the procedural geometric engine in action



STEP 1: GEOMETRIC CHALLENGE

LECCIÓN: 1.1 · Cubo con Ejes X, Y, Z ▾ En curso ★ Nuevo

1.1 Cubo con Ejes X, Y, Z — 3 ejes de apoyo proyectados (Fuga Suave) Ejes X, Y, Z

DESAFÍO #101 30.0s · ARRANCA AL TRAZAR

Deshacer Borrar 0/5 aristas COMPROBAR ✓

STEP 2: FREEHAND SKETCHING

LECCIÓN: 1.1 · Cubo con Ejes X, Y, Z ▾ En curso ★ Nuevo

1.1 Cubo con Ejes X, Y, Z — 3 ejes de apoyo proyectados (Fuga Suave) Ejes X, Y, Z

DESAFÍO #24643 7.3s

Deshacer Borrar 5/5 aristas COMPROBAR ✓

STEP 3: ANALYTICAL EVALUATION

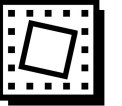
LECCIÓN: 1.1 · Cubo con Ejes X, Y, Z ▾ Superado ✓ Nuevo

1.1 Cubo con Ejes X, Y, Z — 3 ejes de apoyo proyectados (Fuga Suave) Ejes X, Y, Z

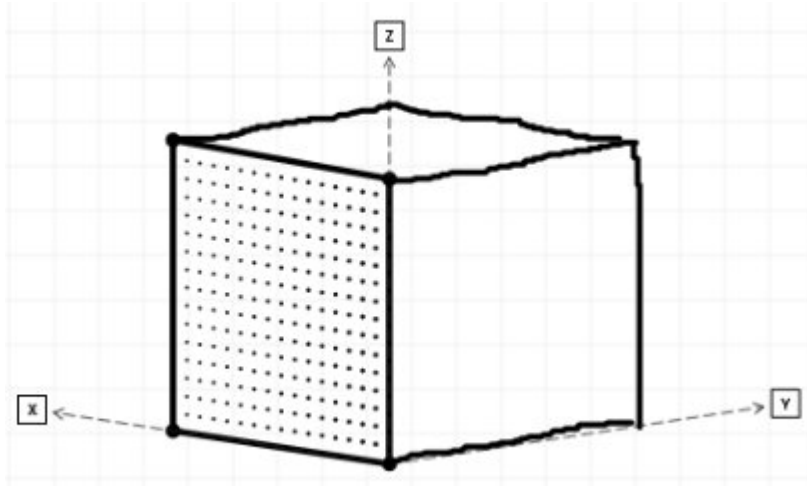
DESAFÍO #24643 0.6s

Deshacer Borrar 5/5 aristas 86% SIGUIENTE →

APROBADO ✓ Cubo aprobado ✓ Buena proporción. Mínimo: 75%
Estructura correcta dentro de los márgenes admisibles de fuga y proporción.



Anatomical Detection of Human Hand Strokes



MULTI-FACTORIAL SCORING FUNCTION

Weighted Analytical Scoring (0% - 100%)

Edge Completion (45%)

Topological volume verification. If an essential edge is missing, the 3D volume does not exist in projective space.

Vanishing Geometric Accuracy (45%)

Dot product between the stroke vector and exact target edge, relative length, and vanishing convergence.

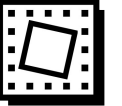
Cleanliness & Stability (10%)

Soft penalty for spurious hand jitter or disconnected phantom lines outside the main volume.

04. METHODOLOGY

Ecosystem Innovations: A4 Worksheets

Bridging the learning experience between tactile graphite and digital feedback



ANALOG-DIGITAL BRIDGE

Vector A4 Worksheets with QR Validation

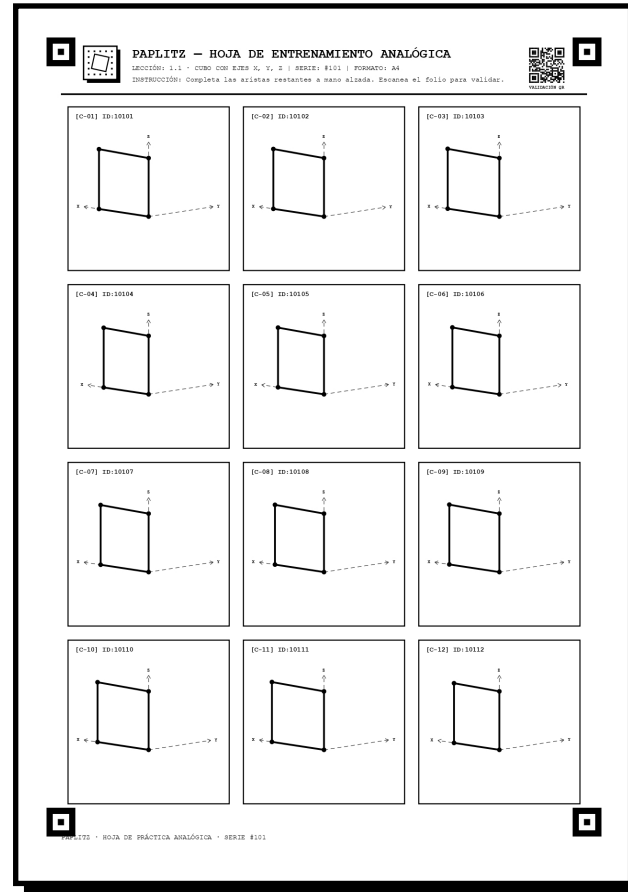
For workshops or students who prefer the physical feel of graphite on traditional paper, Paplitz offers a hybrid workflow.

Worksheet Exercise Generator

Generates printable PDF practice sheets with millimeter calibration grids and 12 procedural geometric challenges.

Optical Camera / Web Scanner

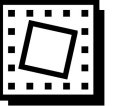
Scan or upload a photo of the completed worksheet; the system decodes the QR code and grades the graphite strokes.



04. METHODOLOGY

Ecosystem Innovations: A4 Worksheets

Bridging the learning experience between tactile graphite and digital feedback



PAPLITZ — HOJA DE ENTRENAMIENTO ANALÓGICA

LECCIÓN: 1.1 · CUBO CON EJES X, Y, Z | SERIE: #101 | FORMATO: A4

INSTRUCCIÓN: Completa las aristas restantes a mano alzada. Escanea el folio para validar.

VALIDACIÓN QR

[C-01] ID:10101

[C-02] ID:10102

[C-03] ID:10103

[C-04] ID:10104

[C-05] ID:10105

[C-06] ID:10106



PAPLITZ — INFORME DE CORRECCIÓN ANALÓGICA A4

LECCIÓN: 1.1 · CUBO CON EJES X, Y, Z | SERIE: #101 | NOTA MEDIA: 92% (12/12 APROBADAS)

EVALUACIÓN: Superposición de la geometría ideal 3D (rojo) sobre tu dibujo escaneado.

VALIDACIÓN QR

[C-01] #10101 97% OK

[C-02] #10102 87% OK

[C-03] #10103 83% OK

[C-04] #10104 89% OK

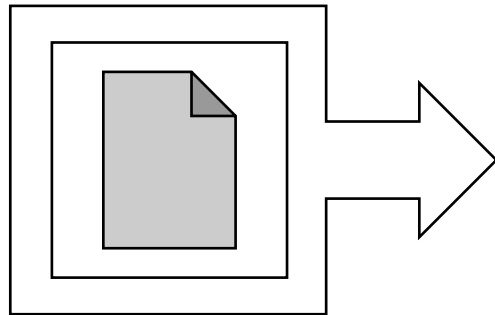
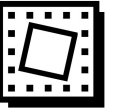
[C-05] #10105 92% OK

[C-06] #10106 94% OK

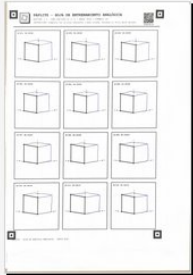
04. METHODOLOGY

Ecosystem Innovations: A4 Worksheets

Bridging the learning experience between tactile graphite and digital feedback



HOJA PROCESADA:



✓ Procesamiento completado con éxito. Nota media: 92% (12/12 aprobadas).  QR LEÍDO **92%**
NOTA MEDIA

Casilla #1 97%	Casilla #2 87%	Casilla #3 89%	Casilla #4 89%
Casilla #5 92%	Casilla #6 96%	Casilla #7 91%	Casilla #8 94%
Casilla #9 94%	Casilla #10 92%	Casilla #11 96%	Casilla #12 91%

 VER HOJA Y CORRECCIONES EN DETALLE

 DESCARGAR PDF

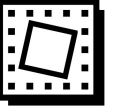


**!Scan and
get your
score!**

04. METHODOLOGY

Gamification & Minigames to Boost Engagement

Introducing motivational game loops for spatial drills





Fiebre de Tiempo

★ NUEVO MODO

CONTRARRELOJ + RECARGA

Empiezas con 25s. Cada cubo aprobado suma **+8s**. ¡Si te toca un **Cubo con Estrella (★)** y lo superas, recargas **+20s**! Deshacer penaliza **-2s** y borrar todo está prohibido.

Récord en 2.1:  0 cubos

▶ JUGAR FIEBRE DE TIEMPO



Cubo Blitz

60 SEGUNDOS PUROS

¿Cuántos cubos bien contruidos eres capaz de clavar en un minuto? Cada aprobado suma **+1 cubo** al instante; los fallos restan 3 segundos de penalización.

Récord en 2.1:  0 cubos

▶ JUGAR BLITZ (60S)



Supervivencia

3 VIDAS MÁXIMAS

3 vidas. Tienes 20 segundos por cada cubo. Cada fallo de perspectiva o línea curva te cuesta un corazón. ¿Cuál es la mayor racha consecutiva que puedes lograr?

Récord en 2.1:  0 seguidos

▶ JUGAR SUPERVIVENCIA



Sprint 5 Cubos

TIME ATTACK

El cronómetro corre hacia arriba: el objetivo es completar 5 cubos aprobados consecutivamente en el menor tiempo humano posible.

Récord en 2.1:  -

▶ JUGAR SPRINT

Play while you
learn!



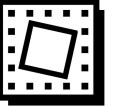
05.

CONCLUSIONS

Present and Future of Paplitz

The philosophy driving the project





OPEN-SOURCE PHILOSOPHY (MIT)

The Universal Right to Master Sketching

- **Educational Democratization:**

Professional design software and educational licenses often carry high costs. Paplitz is 100% free and fully open source under the MIT license.

- **Invitation to Faculty and Students:**

Paplitz is designed to be an evolving living canvas. The goal is for schools of industrial design, architecture, and fine arts to adopt it in their curricula and contribute new challenges.

CONCLUSIONS & CLOSING

The Future of Technical Sketching is Collaborative and Measurable

01

Scientific Pedagogy

Replaces trial-and-error frustration with measurable, objective, and adaptive progress grounded in leading design literature.

02

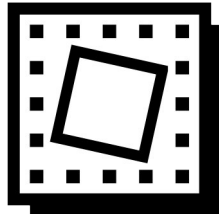
Technological Independence

A pure 3D analytical engine in TypeScript operating client-side at zero operational cost, without external dependencies or latency.

03

Community Impact

An open-source project under MIT license built by and for the design community, immediately available for testing today.



ACCESS THE PROJECT & TRY IT OUT

Live Web App: <https://paplitz.vercel.app/>

Open Repository: <https://github.com/lazaro-guerrero-losada/paplitz>

Contact: Lázaro Guerrero Losada — lazaroguerreroslada@uma.es

Thank You! / Q&A Session